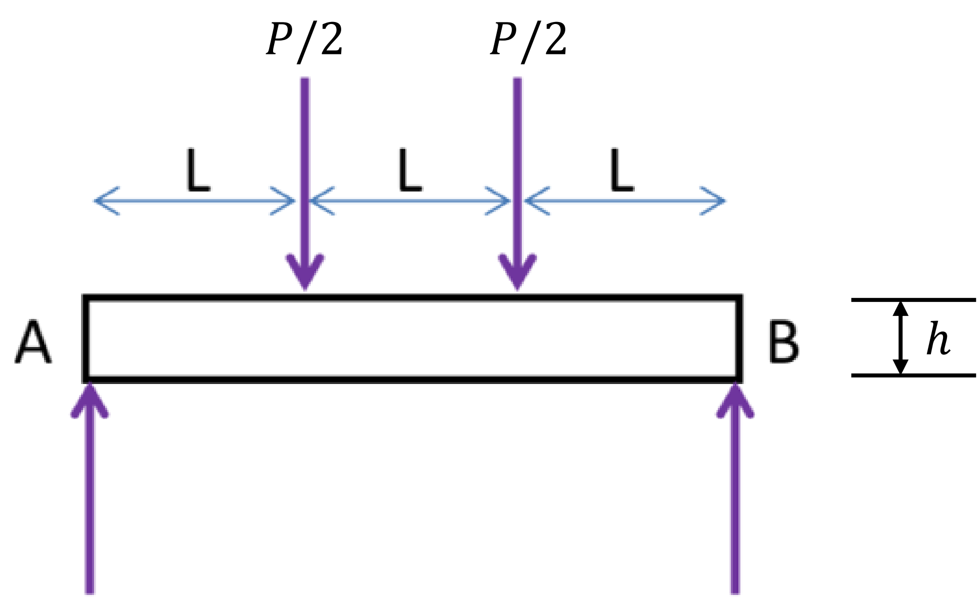
1. The image below shows a schematic for a 4-point bending test. The beam has height $h = 4$ mm and depth $b = 10$ mm (not shown in the figure). The length $L = 30$ mm. Determine the force P and corresponding **normal** strain at the middle point of beam **surface** when the beam starts to yield. Assume the material is carbon and alloy steels (material designation: SAE 5140 OQT 400). Hint: All material properties including modulus of elasticity can be found in Appendix 3 of the textbook. (**10'**)



Solutions to Assignment #1:

From Eq. (2-7) in the textbook, the maximum normal stress in a 4-point bending is

$$\sigma = \frac{3PL}{bh^2}$$

When the material starts to yield, the normal stress should be equal to Yield strength (σ_Y). From Appendix 3, we find $\sigma_Y = 1560$ MPa, then

$$\sigma = \frac{3PL}{bh^2} = \sigma_Y = 1560 \text{ MPa}$$

Solving above equation, we get

$$P = \frac{\sigma_Y bh^2}{3L} = \frac{1560 \times 10^6 \times 0.01 \times 0.004^2}{3 \times 0.03} = \boxed{2773.33 \text{ kN}}$$

The corresponding normal strain at the middle point of beam surface can be obtained by the Hooke's law:

$$\varepsilon = \frac{\sigma}{E} = \frac{1560 \times 10^6}{207 \times 10^9} = \boxed{0.0075 \text{ mm/mm}}$$

2. <u>Composite design:</u> A composite is made from unidirectional strands of fibers in a polyester matrix. If the ultimate tensile strength and modulus of elasticity of a composite should be at least 1.2 GPa and 50 GPa, respectively, what is the requirement of tensile strength and modulus of elasticity of the reinforcement fibers? List all possible reinforcement materials you can use from Table 2-19 in the textbook that fulfill these requirements simultaneously. Assume the volume fraction of fibers is 30%. Use material data from Table 2-19. (**10'**)

Solutions to Assignment #2:

From Eq. (2-10) in the textbook, The ultimate tensile strength is computed as

$$\sigma_{uc} = \sigma_{uf} V_f + \sigma'_m V_m \qquad (1)$$

Assume that the fibers are linearly elastic to failure, then

$$\varepsilon_f = \frac{\sigma_{uf}}{E_f}$$

At this same strain, the stress in the matrix is

$$\sigma'_m = E_m \varepsilon = E_m \varepsilon_f = E_m \frac{\sigma_{uf}}{E_f} \qquad (2)$$

Substituting Eq. (2) into Eq. (1) we get

$$\sigma_{uc} = \sigma_{uf} V_f + E_m \frac{\sigma_{uf}}{E_f} V_m \geq 1.2 \text{ GPa} \qquad (3)$$

Taking $V_f = 30\%$, $V_m = 70\%$, $E_m = 2.76$ GPa, we have

$$\sigma_{uf}\left(0.3 + \frac{0.7 \times 2.76}{E_f}\right) \geq 1.2 \text{ GPa} \qquad (4)$$

Solutions to Assignment #2, cont'd:

The modulus of elasticity of the composite is computed from Eq. (2-12) in the textbook as

$$E_c = E_f V_f + E_m V_m$$

With requirement $E_c \geq 50\ GPa$ and modulus of elasticity of polyester $E_m = 2.76\ GPa$, we have

$$0.3 \times E_f + 0.7 \times 2.76 GPa \geq 50\ GPa$$

$\rightarrow$ $\boxed{E_f \geq 160.23\ GPa}$

Substituting E_f into Eq. (4) we get

$$\boxed{\sigma_{uf} \geq 3.845\ GPa}$$

From Table 2-19 we find that, only **Carbon-PAN (high-strength)** reinforcement material fulfills the above two requirement simultaneously.

3. Shaft design under torsion: A hollow rectangular steel tube (see cross-section in Figure 3-10) has outside dimensions of 4.00 by 2.00 inch and a wall thickness of 0.109 inch. Compute the maximum torque that can be applied to such a tube if the shear stress is to be limited to 6000 psi. For this torque, also compute the angle of twist of the tube over a length of 6.5 ft. Assume the tube material is 18-8 (304) stainless steel with modulus of elasticity 28×10^6 psi. *Hint:* you need to use material property relationship Eq. (2-5) and Poisson's ratio to be found in Table 2-1 in the textbook. **(10')**

Solutions to Assignment #3:

For torsion in members having non-circular cross-sections, we need to find Q and K.

According to Figure 3-10, we have

$a = 2.0$ in; $b = 4.0$ in; $t = 0.109$ in

$$K = \frac{2t(a-t)^2(b-t)^2}{(a+b-2t)} = \frac{2\times 0.109(2-0.109)^2(4-0.109)^2}{(2+4-2\times 0.109)} = 2.0412\ \text{in}^4$$

$$Q = 2t(a-t)(b-t) = 2\times 0.109\times(2-0.109)(4-0.109) = 1.6040\ \text{in}^3$$

Using Eq. (3-12), $\tau_{\max} = T_{\max}/Q$, we have

$$T_{\max} = \tau_{\max} Q = \frac{6000\text{lb}}{\text{in}^2}\times 1.6040\text{in}^3 = \boxed{9624\ \text{lb}\cdot\text{in}}$$

Solutions to Assignment #3, cont'd:

To compute the angle of twist, we need to know shear modulus G.

We first find out the Poisson's ratio (ν) from Table 2-1 in the textbook:

$$\nu = 0.30$$

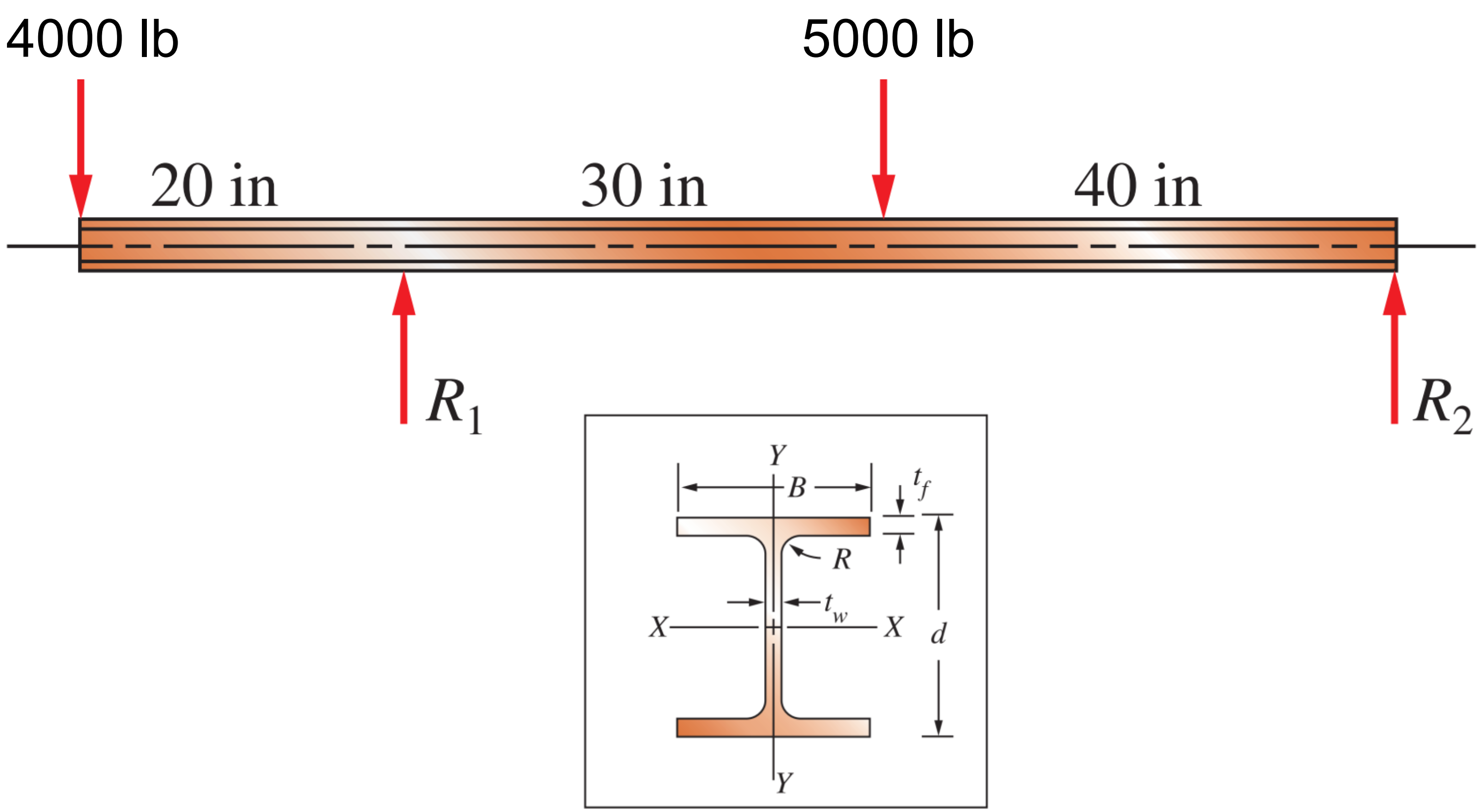
Then, we use material property relationship Eq. (2-5) and we have

$$G = \frac{E}{2(1+\nu)} = \frac{28 \times 10^6}{2(1+0.30)} = 10.7692 \times 10^6 \text{ psi}$$

Using Eq. (3-13), we get the angle of twist

$$\theta = \frac{TL}{GK} = \frac{9624\text{lb} \cdot \text{in} \times 6.5 \times 12\text{in}}{10.7692 \times 10^6 \frac{\text{lb}}{\text{in}^2} \times 2.0412 \text{ in}^4} = \boxed{0.0341 \text{ rad}}$$

4. <u>Beam design under bending:</u> From Table 15-11 in Appendix 15 in the textbook, select a smallest possible cross-section for an aluminum I-beam shape to carry the load shown below with a maximum normal stress of 12,000 psi. The cross-section of the I-beam is shown at the bottom and the bending is along the X-X axis. (**15'**)



Solutions to Assignment #4:

Step 1: Determine the support reactions (R_1 and R_2) for the loadings by using Free-Body-Diagrams:

$$\sum F_y = 0 \quad \rightarrow \quad R_1 + R_2 = 4000 + 5000 = 9000 \text{ lb} \qquad (1)$$

$$\sum M_{R_1} = 0 \quad \rightarrow \quad 4000 \times 20 + R_2 \times 70 = 5000 \times 30 \text{ lb} \cdot \text{in} \qquad (2)$$

Solving Eqs. (1) and (2) we get:

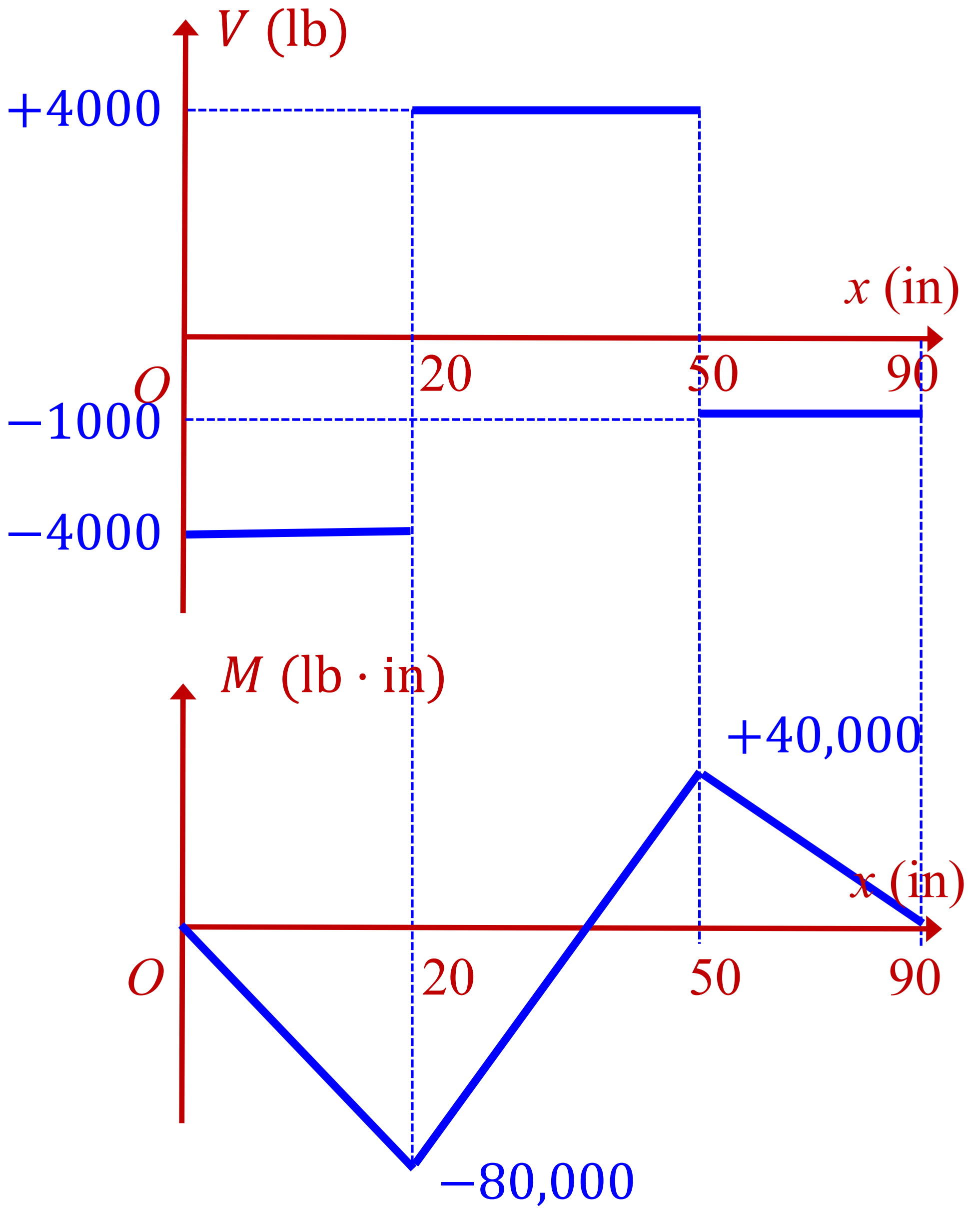
$R_1 = 8000$ lb

$R_2 = 1000$ lb

Step 2: plot the shear force and bending moment diagrams using graphical method.

From bending moment diagram, we find that the maximum bending moment happens at R_1 position with magnitude of 80,000 lb $\cdot$ in.

Solutions to Assignment #4, cont'd:



Solutions to Assignment #4, cont'd:

Step 3: To meet design stress $\sigma_{\max} = 12{,}000$ psi, the required section modulus should be at least:

$$S \geq \frac{M_{\max}}{\sigma_{\max}} = \frac{80000 \text{ lb} \cdot \text{in}}{12000 \text{ psi}} = 6.67 \text{ in}^3$$

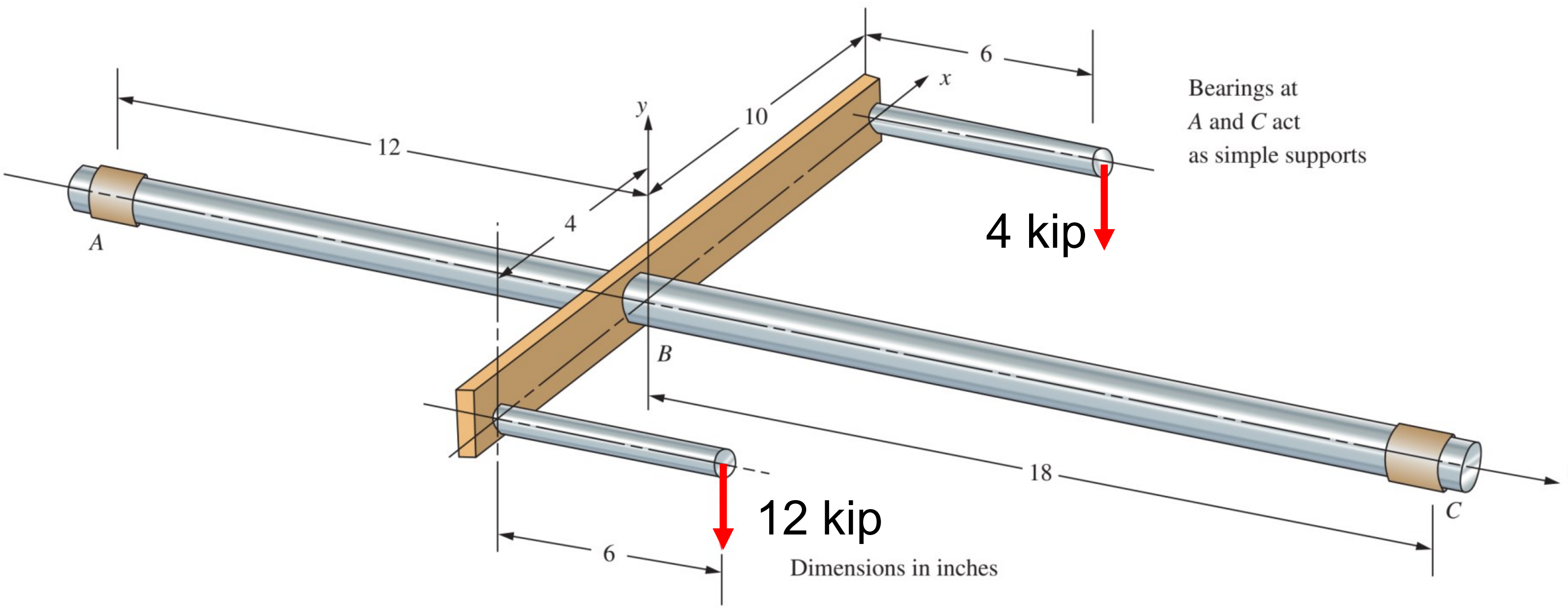
Step 4: Reading Appendix 15 – 11 we can select:

Depth 6 in, width 4 in, with $S = 7.33 \text{ in}^3$.

TABLE 15–11 I-Beam shapes: Aluminum, Aluminum Association Standard Shapes, Larger Sizes: 3 in to 12 in Depth

Size							[1]Section properties					
							Axis X–X			Axis Y–Y		
Dept d (in)	Width B (in)	Area A (in^2)	[2]Weight (lb/ft)	Flange thickness t_f (in)	Web thickness t_w (in)	Fillet radius R (in)	I_x (in^4)	S_x (in^3)	r_x (in)	I_y (in^4)	S_y (in^3)	r_y (in)
3.00	2.50	1.392	1.637	0.20	0.13	0.25	2.24	1.49	1.27	0.52	0.42	0.61
3.00	2.50	1.726	2.030	0.26	0.15	0.25	2.71	1.81	1.25	0.68	0.54	0.63
4.00	3.00	1.965	2.311	0.23	0.15	0.25	5.62	2.81	1.69	1.04	0.69	0.73
4.00	3.00	2.375	2.793	0.29	0.17	0.25	6.71	3.36	1.68	1.31	0.87	0.74
5.00	3.50	3.146	3.700	0.32	0.19	0.30	13.94	5.58	2.11	2.29	1.31	0.85
6.00	4.00	3.427	4.030	0.29	0.19	0.30	21.99	7.33	2.53	3.10	1.55	0.95

5. <u>Beam design under combined loadings 1:</u> Determine the minimum diameter of the main shaft portion, labeled A, B, and C, by considering both shear and tensile strength. Include any unbalanced torque on the shaft that tends to rotate it about the z-axis. Draw the complete shearing force and bending moment diagrams for loading in the y–z plane. The shaft material is Aluminum 7075-T6. Find the material property in Appendix in the textbook. **(15')**



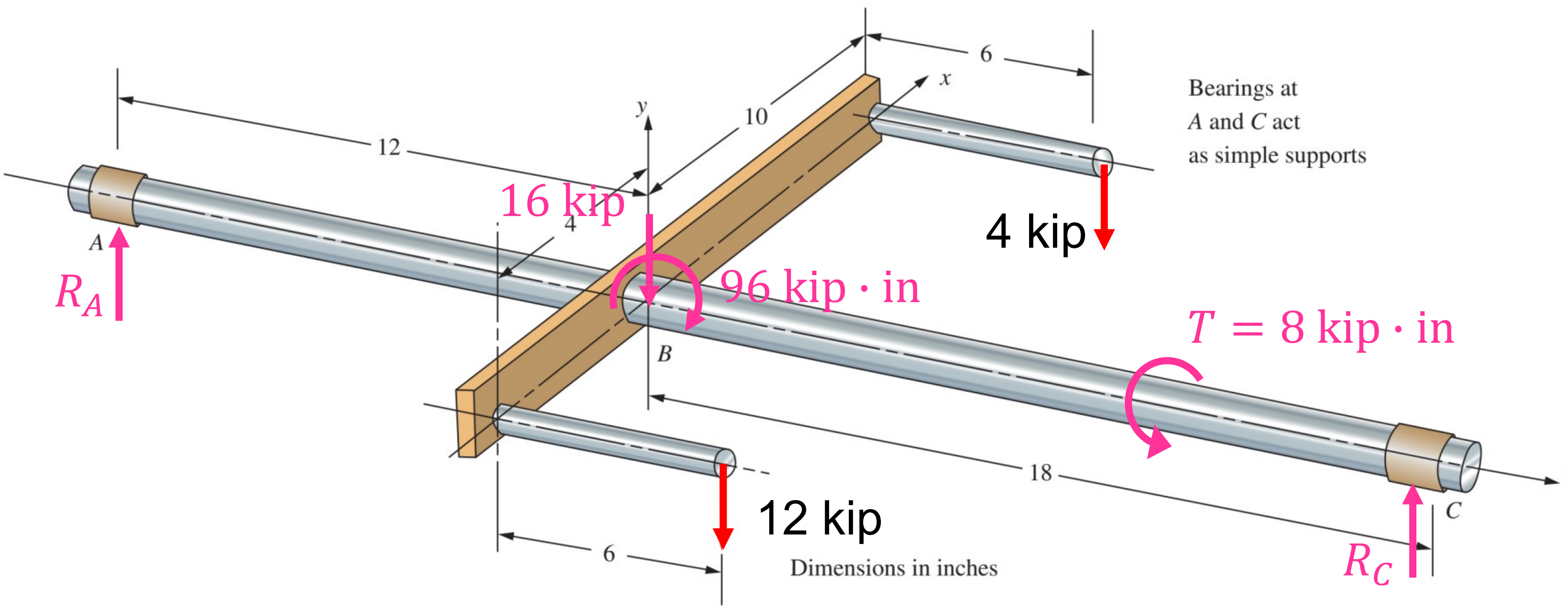
Solutions to Assignment #5:

Step 1: Solve reactions by Free-Body-Diagram analysis:

$$R_A + R_C = 12 + 4 = 16 \text{ kip} \quad (1)$$
$$R_C \times (18 + 12) = 16 \times 12 + 96 = 288 \text{ kip} \cdot \text{in} \quad (2)$$

Solving Eqs. (1) and (2) we get the reaction forces:

$$R_A = 6.4 \text{ kip}, \qquad R_C = 9.6 \text{ kip}$$

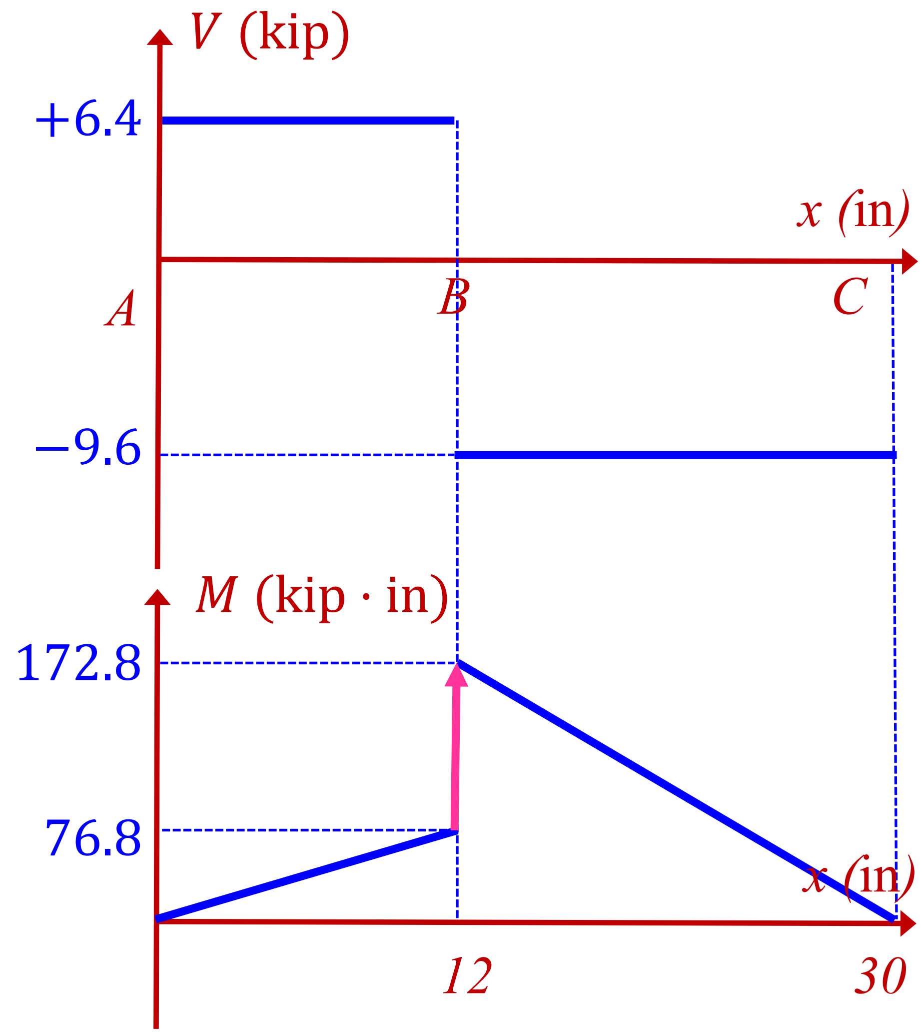


Bearings at A and C act as simple supports

Solutions to Assignment #5, cont'd:

Step 2: Plot shear force and bending moment diagrams with graphical method. With moment diagram, we find the maximum bending moment happens at point B with magnitude

$$M_{\max} = 172.8 \text{ kip} \cdot \text{in}$$



Solutions to Assignment #5, cont'd:

Step 3: For Aluminum 7075-T6, tensile strength is 83 ksi (**Appendix 9**). To meet design stress $\sigma_\mathrm{d} = 83$ ksi, the required section modulus for bending should be

$$S \geq \frac{M_\mathrm{max}}{\sigma_\mathrm{d}} = \frac{172.8 \text{ kip}\cdot\text{in}}{83 \text{ ksi}} = 2.0819 \text{ in}^3 \qquad (3)$$

From Appendix 1, for circular cross-section, the section modulus of bending is

$$S = \frac{\pi D^3}{32} \qquad (4)$$

Combining Eqs. (3) and (4), we determine that, to meet the design criterion for *bending*, the diameter should be

$$D \geq 2.768 \text{ in} \qquad (5)$$

Solutions to Assignment #5, cont'd:

Step 4: There is a torque applied to the shaft with magnitude:

$$T = 12 \times 4 - 4 \times 10 = 8 \text{ kip} \cdot \text{in} \qquad (6)$$

From Appendix 1, for circular cross-section, the polar section modulus is

$$Z_p = \frac{\pi D^3}{16} \qquad (7)$$

For Aluminum 7075-T6, shearing strength is 48 ksi (**Appendix 9**). Thus,

$$\tau_{\max} = \frac{T}{Z_p} \le 48 \text{ ksi}$$

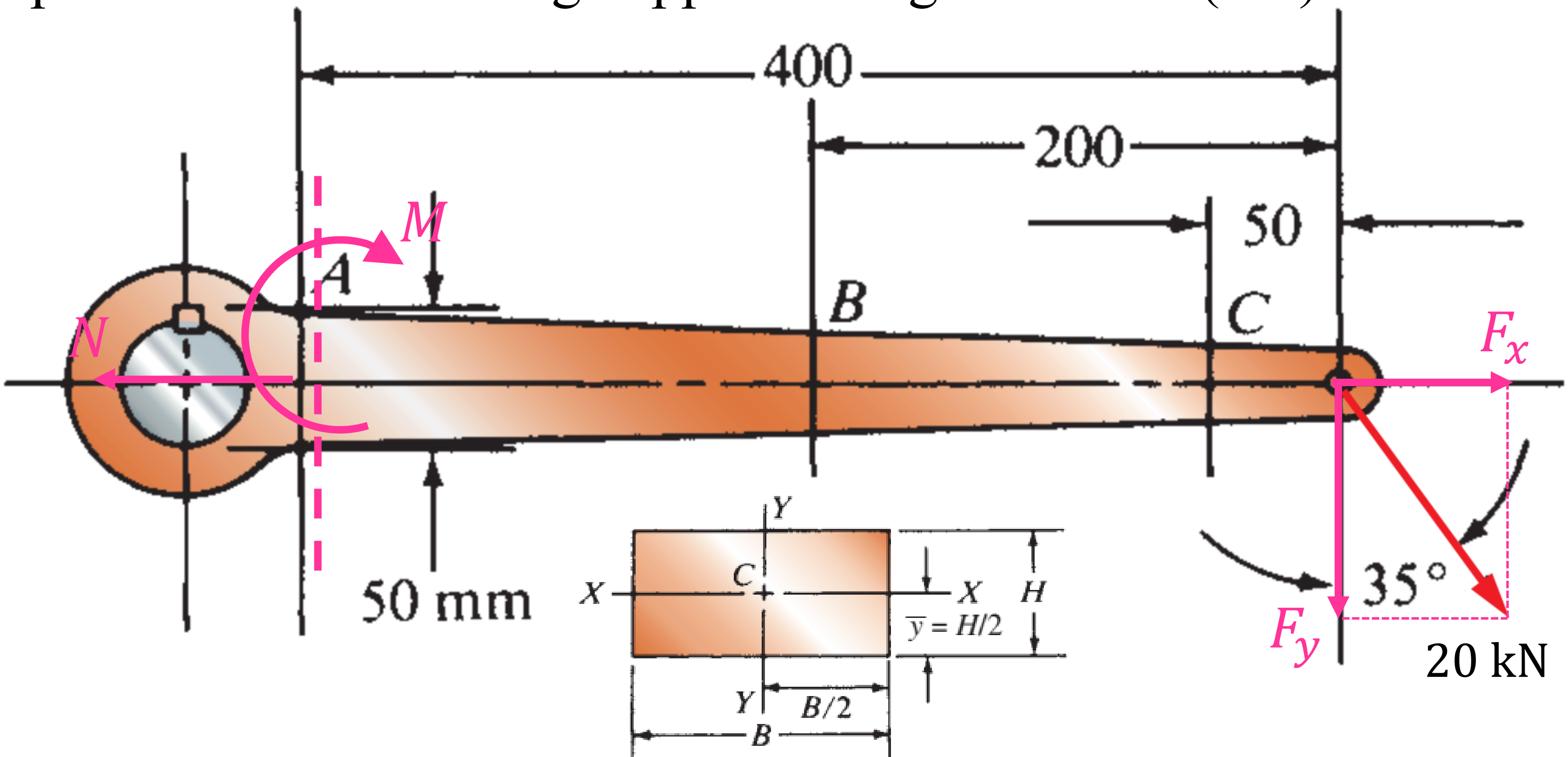
Then, to meet the design criterion for *torsion*, the diameter should be

$$\frac{8 \text{ kip} \cdot \text{in}}{\frac{\pi D^3}{16}} \le 48 \text{ ksi}$$

$$D \ge 0.9468 \text{ in} \qquad (8)$$

Combining (5) and (8), finally, we determine that, to meet design criteria for both bending and torsion, the minimum diameter of the shaft should be $\boxed{2.768 \text{ in.}}$

6. <u>Beam design under combined loadings 2:</u> Determine the minimum thickness of the lever at position A, by only considering the tensile strength of the lever. The lever material is Cast Iron Gray iron-ASTM A48 No. 60A. Find the standard value of the ultimate strength (σ_u) in Appendix in the textbook. The cross-section is rectangular with height $H = 50$ mm at position A and bending happens along XX-axis. **(10')**



Solutions to Assignment #6

Step 1: The external force can be decomposed into two components in x- and y-direction, respectively:

$F_x = 20 \times \sin 35° = 11.47 \text{ kN}$

$F_y = 20 \times \cos 35° = 16.38 \text{ kN}$

Step 2: Using section method at position A and applying Free-Body-Diagram, we can get the internal resultant tensile force and bending moment:

$N = F_x = 11.47 \text{ kN}$

$M = F_y \times 0.4\text{m} = 16.38\text{N} \times 0.4\text{m} = 6.552 \text{ kN} \cdot \text{m}$

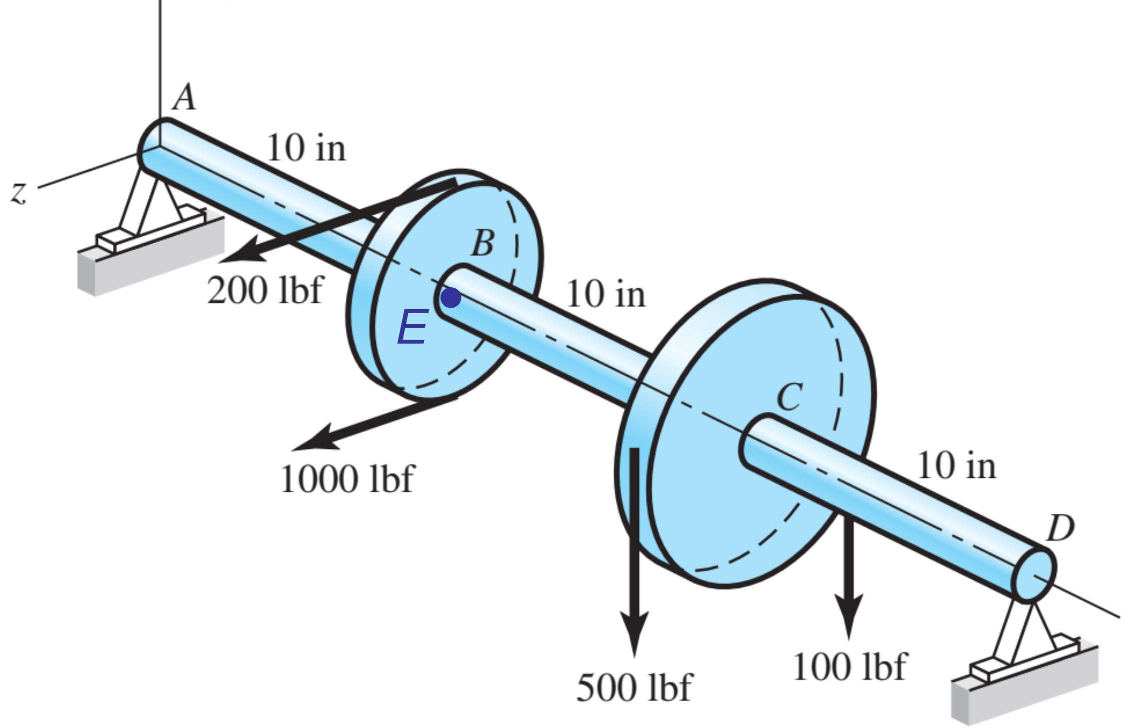
Step 3: The maximum tensile stress is composed of both the internal resultant tensile force and the tensile stress induced by bending moment (refer to Appendix 1 in textbook for section modulus and **Appendix 8** for ultimate strength for Gray iron-ASTM A48 No. 60A):

$$\sigma_{\max}|_A = \frac{N}{A} + \frac{M}{S_A} = \frac{11.47 \times 10^3}{0.05B} + \frac{6.552 \times 10^3}{0.05^2 B/6} \leq 414 \times 10^6 \text{ Pa}$$

Find ultimate strength from **Appendix 8**

$\rightarrow \quad B \geq 0.0385 \text{ m} \;$ or $B \geq \boxed{38.5 \text{ mm}}$

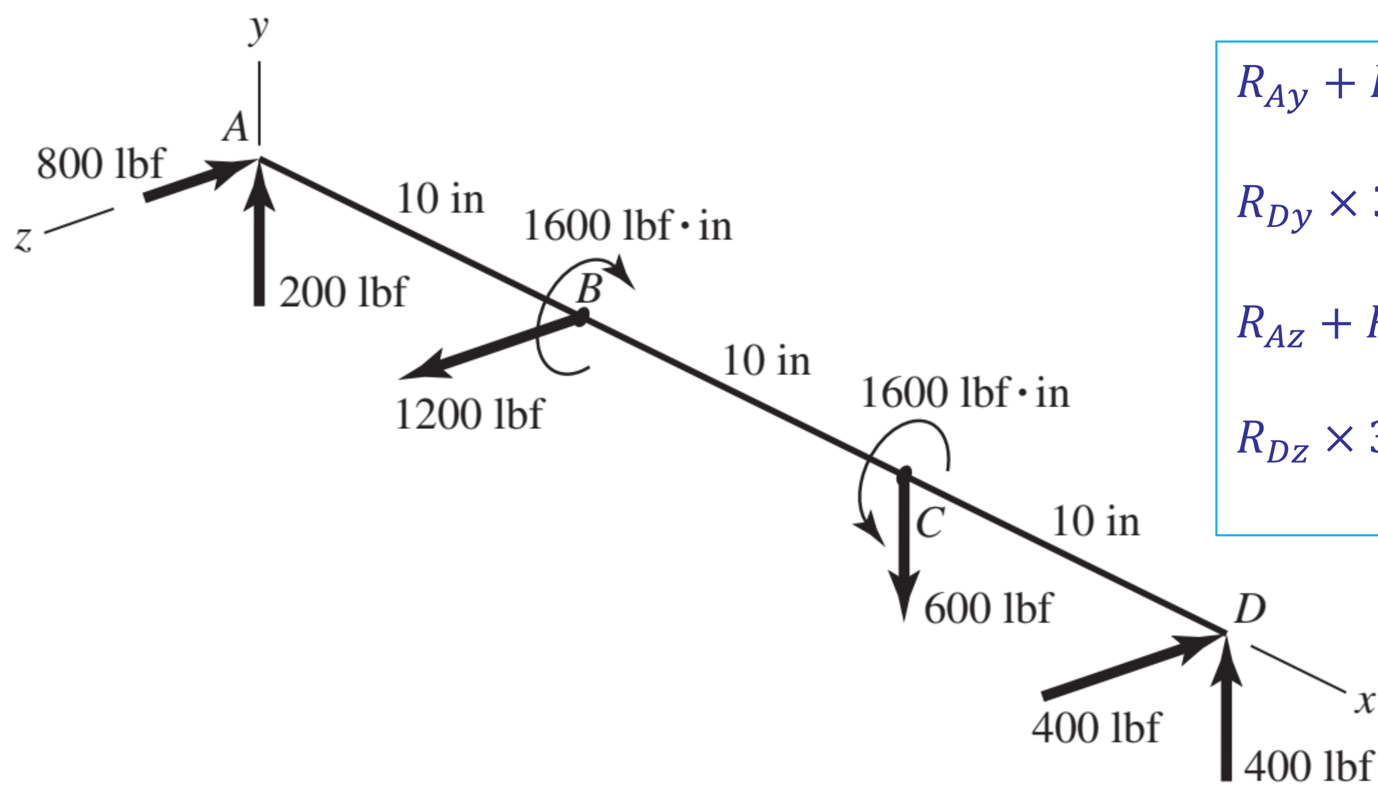
7. <u>Mohr's circle for combined loadings 1:</u> A solid steel shaft with diameter of 1.5 in is simply supported at the ends. Two pulleys are keyed to the shaft where pulley B is of diameter 4.0 in and pulley C is of diameter 8.0 in. Considering bending and torsional stresses only, determine the maximum tensile stress and absolute maximum shear stress on a surface point E just right to the point B. Hint: The net maximum moment does not necessarily occur along y- or z-direction. **(15')**



Solutions to Assignment #7

Step 1: We show the net forces, reactions, and torsional moments on the shaft. Although this is a three-dimensional problem and vectors might seem appropriate, we will look at the components of the moment vector by performing a two-plane analysis.

The reactions at point A and D can be solved by using Free-Body-Diagrams.



$$R_{Ay} + R_{Dy} = 600$$

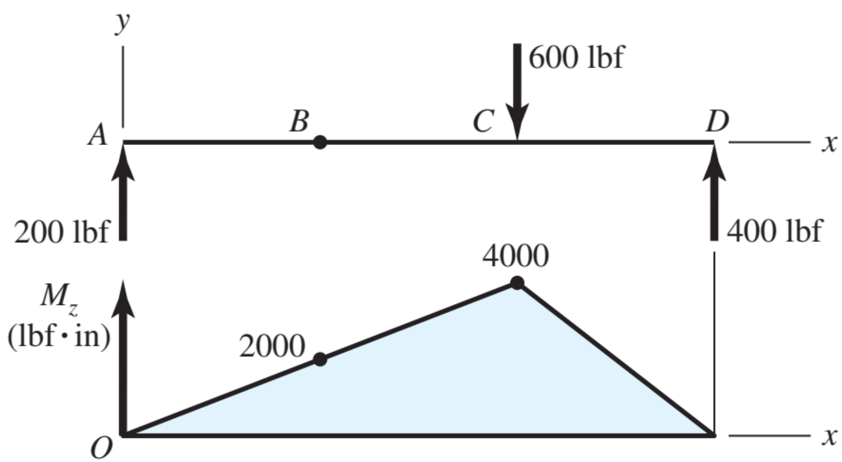
$$R_{Dy} \times 30 = 600 \times 20$$

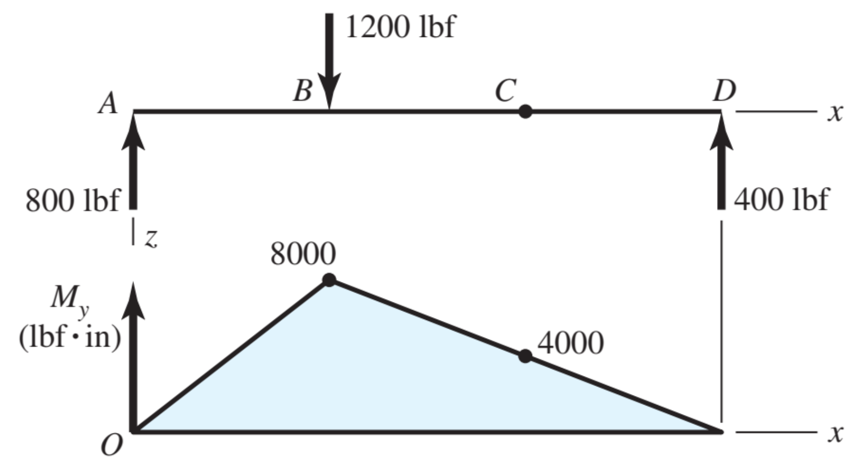
$$R_{Az} + R_{Dz} = 1200$$

$$R_{Dz} \times 30 = 1200 \times 10$$

Solutions to Assignment #7, cont'd

Step 2: We show the loading in the xy plane, as viewed down the z axis, where bending moments are actually vectors in the z direction. Thus, we label the moment diagram as M_z versus x as shown on the left. For the xz plane, we look down the y axis, and the moment diagram is M_y versus x as shown on the right.





The net moment on a section is the vector sum of the components. That is,

$$M = \sqrt{M_y^2 + M_z^2}$$

At point B, $M_B = \sqrt{2000^2 + 8000^2} = 8246 \text{ lb} \cdot \text{in}$

Solutions to Assignment #7, cont'd

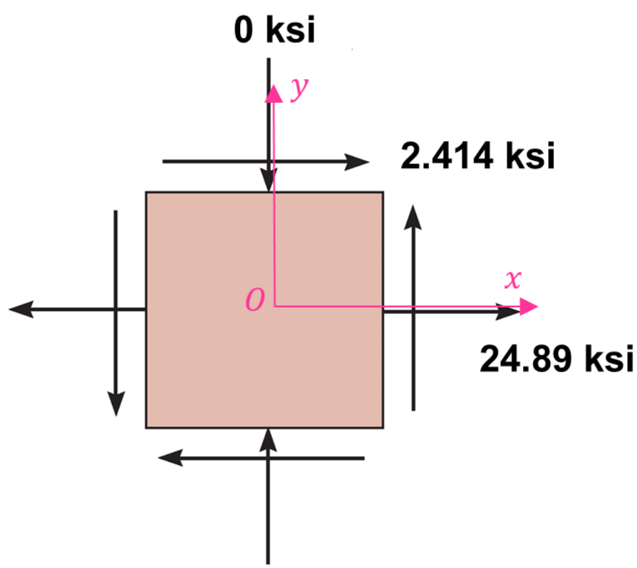
Step 3: The maximum tensile stress due to bending moment at a surface point E just right to point B is

$$\sigma = \frac{M_B}{S} = \frac{M_B}{\pi D^3/32} = \frac{8246}{\pi 1.5^3/32} = 24.89 \text{ ksi}$$

The maximum shear stress due to torsion at the same surface point E is

$$\tau = \frac{T}{Z_p} = \frac{T}{\pi D^3/16} = \frac{1600}{\pi 1.5^3/16} = 2.414 \text{ ksi}$$

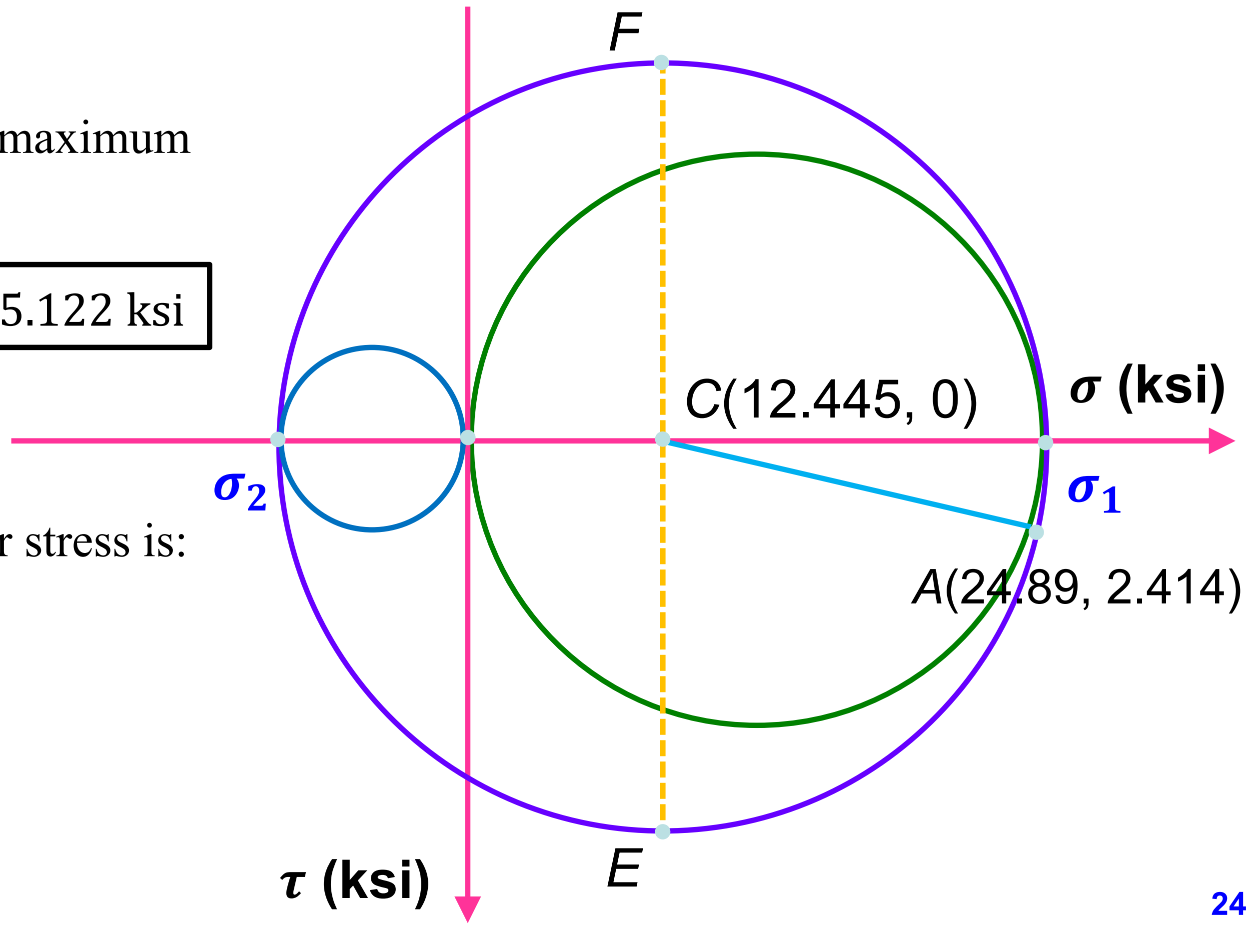
Step 4: Use the tensile stress and shear stress component to plot the stress state (stress square).



Solutions to Assignment #7, cont'd

Step 5: Given $\sigma_x = +24.89$ ksi, $\sigma_y = 0$, $\tau_{xy} = +2.414$ ksi, the center of the circle C is located on the σ-axis, at the point $\sigma_{\text{avg}} = \frac{\sigma_x+\sigma_y}{2} = 12.445$ ksi

The point $C(12.445, 0)$ and the reference point $A(24.89, 2.414)$ are labeled. Then, the radius of the circle is $R = \sqrt{(12.445 - 24.89)^2 + 2.414^2} = 12.677$ ksi, and the circle is constructed.



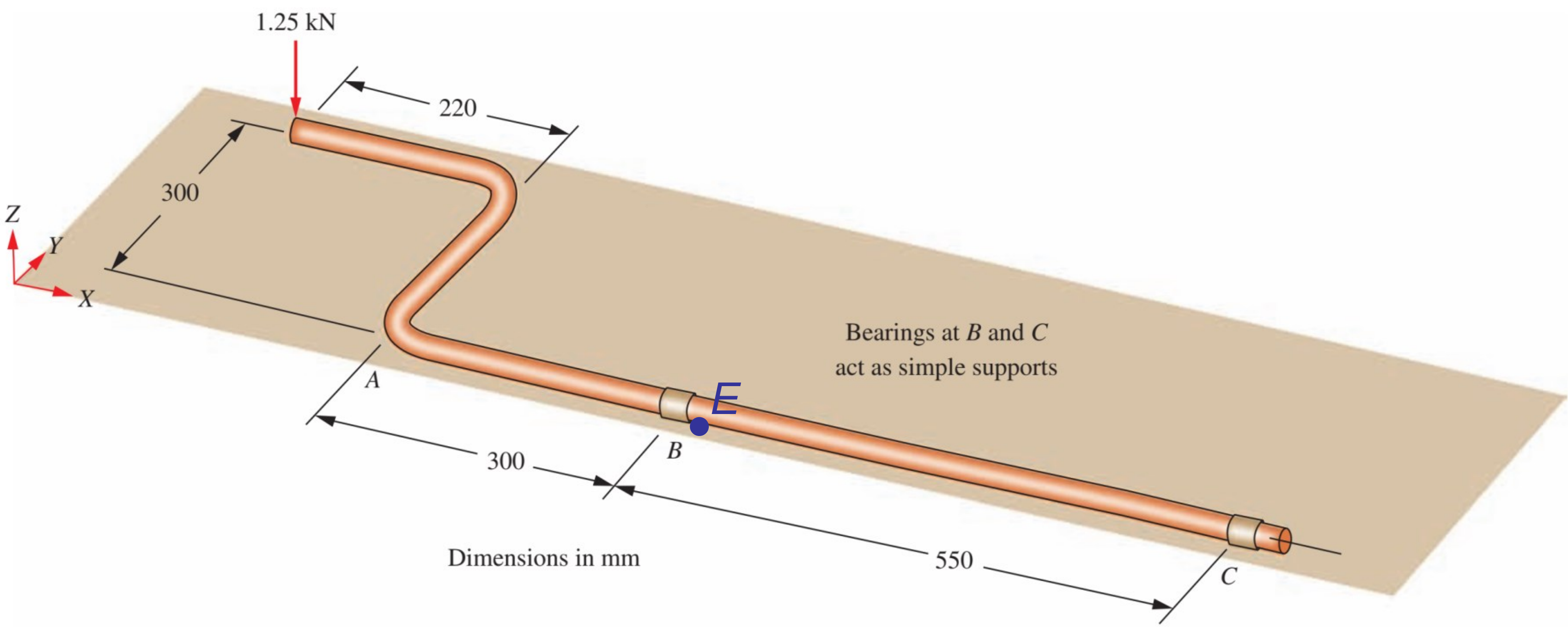
Using the Mohr's circle, the maximum tensile stress is:

$$\sigma_1 = 12.445 + 12.677 = \boxed{25.122 \text{ ksi}}$$

The absolute maximum shear stress is:

$$\tau_{max}^{abs} = R = \boxed{12.677 \text{ ksi}}$$

8. <u>Mohr's circle for combined loadings 2:</u> For the shaft ABC, draw the Mohr's circle to determine the principal stresses and absolute maximum shear stress for a point on the bottom of the shaft just to the right of section B. The torque applied to the shaft at B is resisted at support C only. The shaft diameter is 38 mm. (**15'**)



Solutions to Assignment #8

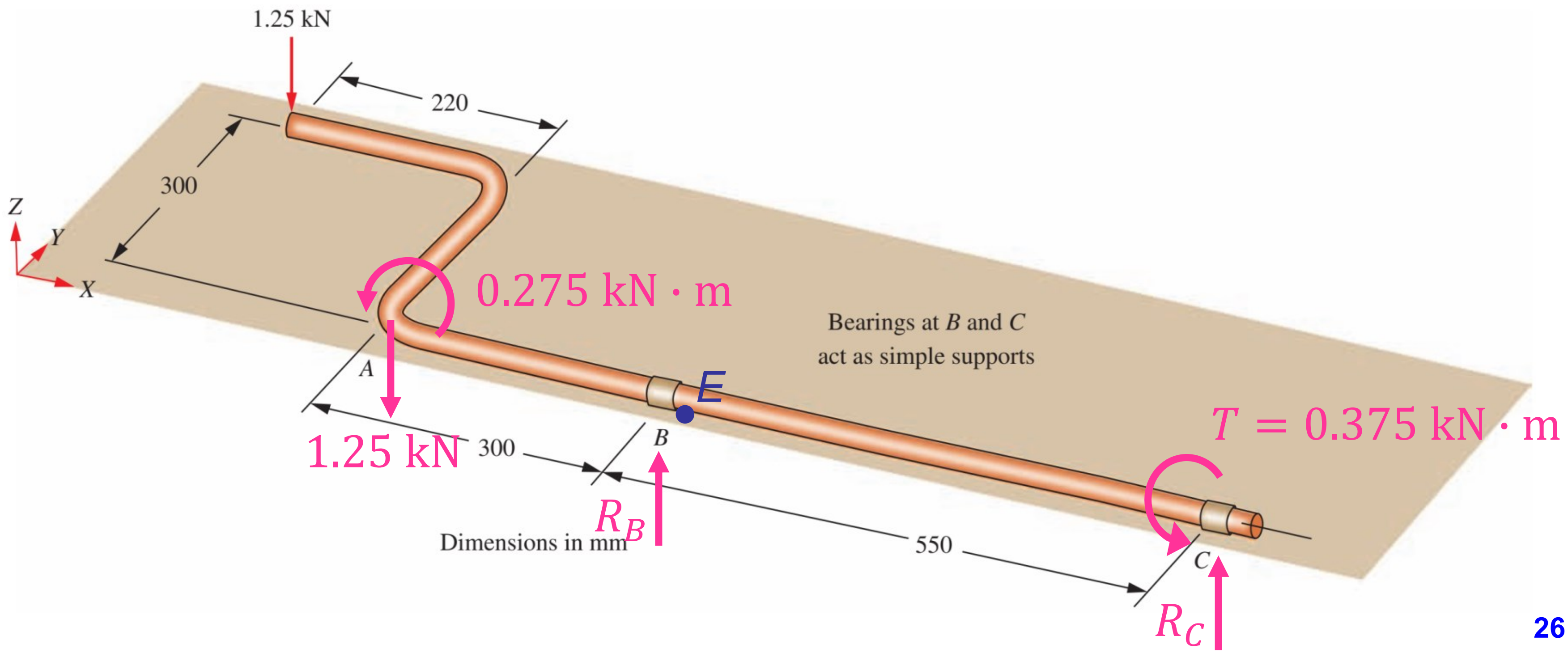
Step 1: We move the equivalent force, bending moment, and torsional moments on the shaft ABC. The reactions at point B and C can be solved by using Free-Body-Diagrams:

$\sum F_z = 0 \quad \rightarrow \quad R_B + R_C = 1.25 \quad (1)$

$\sum M_B = 0 \quad \rightarrow \quad R_C \times 0.55 + 1.25 \times 0.3 + 0.275 = 0 \quad (2)$

Solving Eqs. (1) and (2) we get:

$R_B = 2.43$ kN $\quad R_C = -1.18$ kN (negative sign means the presumed direction should be reversed)



Solutions to Assignment #8, cont'd

Step 2: Plot shear force and bending moment diagrams with graphical method. With moment diagram, we find the bending moment at a point just to the right of section B is

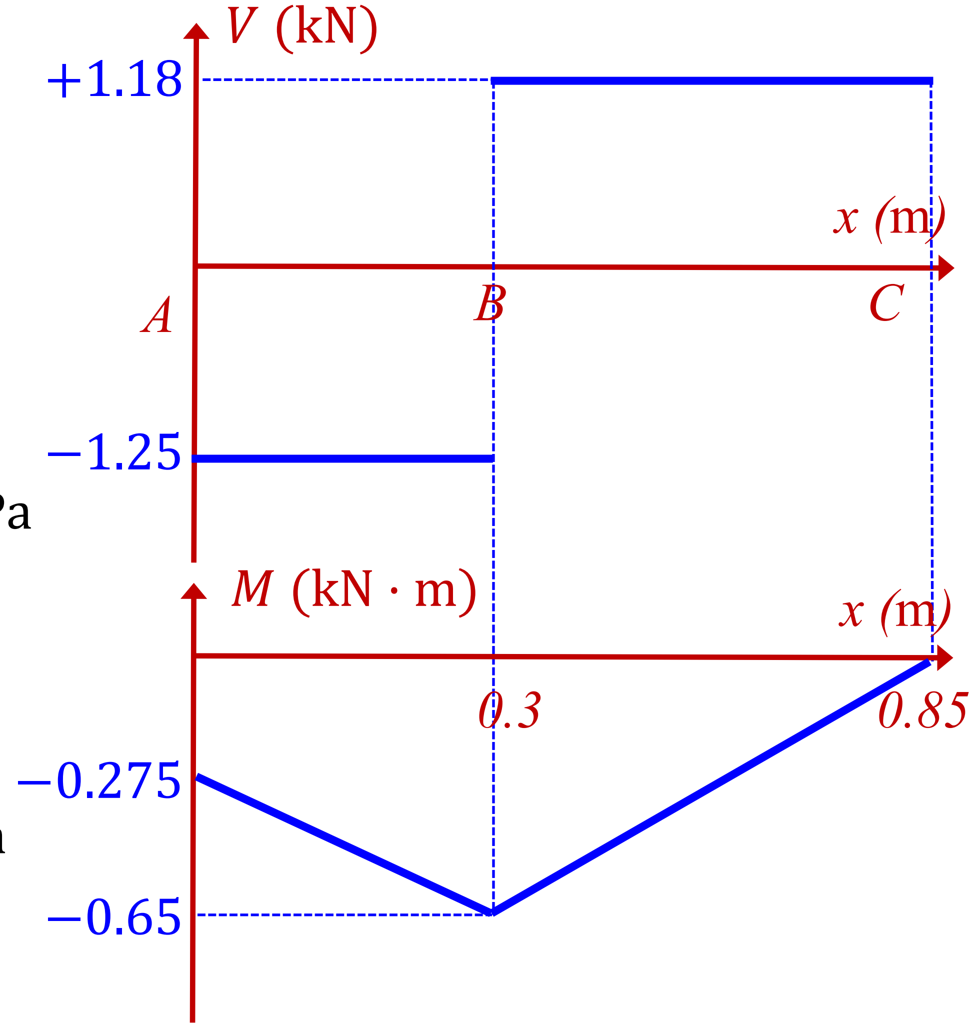
$$M = -0.65\ \text{kN}\cdot\text{m}$$

The corresponding compressive normal stress is

$$\sigma = \frac{M}{S} = \frac{650}{\pi 0.038^3/32} = 120.66\ \text{MPa}$$

Shear stress induced by torque is

$$\tau = \frac{T}{Z_p} = \frac{375}{\pi 0.038^3/16} = 34.81\ \text{MPa}$$



Solutions to Assignment #8, cont'd

Step 3: For a surface point, the stress state is plane stress, with stress components $\sigma_x = -120.66$ MPa, $\sigma_y = 0$, and $\tau_{xy} = 34.81$ MPa.

Then, we can plot the Mohr's circle with center C and radius R.

Center $C\left(\frac{\sigma_x+\sigma_y}{2}, 0\right) = C(-60.33{,}0)$

$$R = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} = \sqrt{60.33^2 + 34.81^2} = 69.65 \text{ MPa}$$

According to the Mohr's circle, the principal stresses are

$$\sigma_1 = \sigma_C + R = -60.33 + 69.65 = \boxed{9.32 \text{ MPa}}$$

$$\sigma_2 = \sigma_C - R = -60.33 - 69.65 = \boxed{-129.98 \text{ MPa}}$$

By plotting 3 Mohr's circles we know the absolute maximum shear stress is

$$\tau_{\max}^{\text{abs}} = R = \boxed{69.65 \text{ MPa}}$$

Solutions to Assignment #8, cont'd